

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5CJ302			
Course Title	ABSTRACT ALGEBRA I			
Type of Course	Major			
Semester	V			
Academic Level	300-399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic set theory, algebra of Integers, operations on functions, basic proof techniques etc.			
Course Summary	This course explores the algebraic concepts of Binary Operations, Binary Structures, Groups, Rings, Integral Domains and Fields. We further study the Theory of Groups. Elementary properties, Subgroups, Finite Groups, Cyclic Groups, Groups of Permutations, Orbits, Cycles, Alternating Groups, Cosets and the Theorem of Lagrange are studied. Then we study mappings between groups or Homomorphisms. Finally, the Open-ended section points to Generating sets, Factor Groups and Field of Quotients of an Integral Domain.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Discuss about binary operations, isomorphic binary structures and groups	U	C	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO2	Analyse and classify subgroups and cyclic groups, and determine their properties using group theory.	An	P	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO3	Evaluate and apply theorems related to cosets, Lagrange's theorem, homomorphisms, rings, and fields to solve complex algebraic problems.	E	F	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text book	A first course in abstract algebra, Fraleigh, John B.. Seventh Edition, Pearson Education India, 2003			
Module	Unit	Content	Hrs (48+12)	Marks Ext(70)
I	Module I		12	Min.15
	1	Section 2- Binary Operations (2.1 to 2.10)		
	2	Section 2- Binary Operations (2.11 to 2.25)		
	3	Section 3- Isomorphic Binary Structures (3.1 to 3.11).		
	4	Section 3- Isomorphic Binary Structures (3.12 to 3.17)		
	5	Section 4- Groups (4.1 to 4.14)		
	6	Section 4- Groups – Elementary Properties of Groups, Finite Groups and Group tables (4.15 onwards)		
II	Module II		14	Min.15
	7	Section 5- Subgroups (5.1 to 5.16)		
	8	Section 5 -Subgroup - Cyclic Subgroups (5.17 to 5.23)		
	9	Section 6 -Cyclic Groups (6.1 to 6.9) (Proof of Theorem 6.3 is optional)		
	10	Section 6- Cyclic Groups (6.10 to 6.17) (Proof of Theorem 6.14 is optional).1		
	11	Section 8-Groups of Permutations (up to 8.6)		
	12	Section 8- Groups of Permutations (8.7 to 8.18)		
III	Module III		10	Min.15
	13	Section 9 - Orbits, Cycles, and the Alternating Groups (Up to 9.10)		
	14	Section 9 - Orbits, Cycles, and the Alternating Groups (9.11 to 9.21) (Proof 2 of theorem 9.15 is optional).		
	15	Section 10- Cosets and the theorem of Lagrange (Up to 10.9)		
	16	Section 10- Cosets and the theorem of Lagrange (10.10 to 10.14)		

IV	Module IV		12	Min.15
	17	Section 13- Homomorphisms (13.1 to 13.10)		
	18	Section 13-Homomorphism (13.11 to 13.20)		
	19	Section 18-Rings and Fields (18.1 to 18.13)		
	20	Section 18-Rings and Fields (18.14 to 18.18)		
	21	Section 19-Integral Domains (19.1 to 19.8)		
	22	Section 19-Integral Domains (19.9 to 19.15)		
V	Module V (Open Ended)		12	-
		Generating Sets in Groups		
		Factor Groups		
		The Field of Quotients of an Integral Domain		

References

1. Herstein, Israel Nathan. *Topics in algebra*. John Wiley & Sons, 1991.
2. Gallian, Joseph. *Contemporary abstract algebra*. Chapman and Hall/CRC, 2021.
3. Wallace, David AR. *Groups, rings and fields*. Springer Science & Business Media, 2001
4. Reis, Clive. *Abstract algebra: an introduction to groups, rings and fields*. World Scientific Publishing Company, 2011.
5. Allan Clark, *Elements of Abstract Algebra*, Dover Publications, 1984
6. C Musili, *Introduction to Rings and Modules*, Narosa Publications, 2009

Suggested Programming Exercises for Open-Ended

1. Form congruence groups, their Cayley tables (Section 9.2, Ref (3)).
2. Form symmetric groups of various orders, list the elements, find the power of some elements, find out the product of some of the elements. Find the order of the elements. Form a group table using conditionals and loops. (Section 9.3, Ref (3) or Ref (1)).
3. List S_3 . Find a subgroup from this group. How many distinct subgroups can be found from this group? List all of them.
4. Form the Dihedral group D_4 , check if it is abelian using `is_abelian()`. Conduct the same experiments as listing the elements ,finding the orders etc as above. (Section 9.4, Ref (3) or Ref (1)).
5. Test the command `is normal ()` on a few subgroups of S_3 . (Ref (1)).
6. Create cyclic groups. (Section 9.5, Ref (3)).

7. Form finitely generated abelian groups. (Section 9.6, Ref (3)).
8. Form a subgroup of a group (say, S_3) (Section 9.8, Ref (3)).

References

1. Robert A. Beezer; Group Theory and SAGE: A Primer, <http://people.reed.edu/~davidp/332/sage-group-theory.pdf>
2. Group Theory and Sage - SageMath tutorial https://doc.sagemath.org/html/en/thematic_tutorials/group_theory.html
3. Ajit Kumar, Vikas Bist; Group Theory An Expedition with SageMath, Narosa Publishing House.
4. Thomas W. Judson, Robert A. Beezer; Abstract Algebra Theory and Applications with Sage Exercises for Abstract Algebra, <http://abstract.ups.edu/download/aata-20130816.pdf>

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	1	0	2	0	0	0	2	0	0
CO 2	1	2	3	0	2	0	2	0	3	0	0
CO 3	0	1	2	3	2	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓